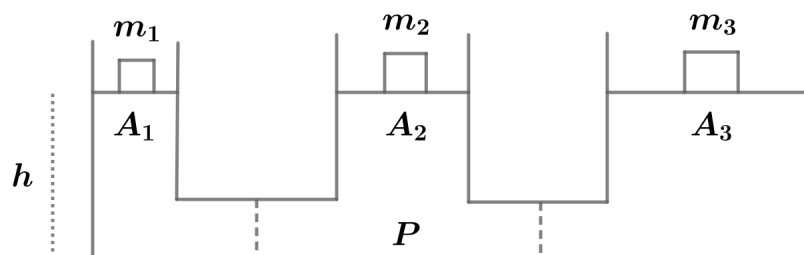


2020B F=ma Exam: Problem 15

Kevin S. Huang



The system is in equilibrium so the pressure is the same everywhere at the bottom:

$$P = P_{atm} + \rho gh + \frac{m_i g}{A_i}$$

Thus,

$$\frac{m_1}{A_1} = \frac{m_2}{A_2} = \frac{m_3}{A_3}$$

Closing either valve doesn't change the pressure so it will have no effect on the height of m_2 or any of the masses. Thus, the answer is A.